

Alzheimer's Disease — The Full Picture

What Alzheimer's actually is

Alzheimer's is a progressive neurodegenerative disease. It is not a normal part of aging. The brain physically shrinks. Neurons die and do not regenerate. It is the most prevalent type of dementia, characterized by pathological deposition of amyloid- β plaques and tau tangles within the brain parenchyma. Synaptic dysfunction occurs at the early stages and then progresses into neurodegeneration with neuronal cell death in later stages. [Nature](#) By the time someone shows symptoms, damage has been happening in the brain for 20–30 years already.

Amyloid-beta — the first domino

Every brain naturally produces a protein called amyloid precursor protein (APP). When APP gets cut by enzymes, it generates small fragments. The problematic one is called amyloid-beta 42 (A β 42). Physiologically, these proteins are produced and expressed within the normal human body. However, under pathological conditions, abnormal expression, post-translational modifications, conformational changes, and truncation can make these proteins prone to aggregation, triggering specific disease-related cascades. [Lippincott Williams & Wilkins](#)

When A β 42 accumulates faster than the brain can clear it, it clumps together. First into small toxic clusters called oligomers — these are the most damaging form; they directly poison synapses. Then into larger insoluble plaques that sit between neurons like hardened concrete. The brain's immune cells (microglia) are supposed to clean this up. When they can't keep pace, the plaques build up and the inflammatory response never turns off.

The "amyloid cascade hypothesis" — the idea that amyloid buildup is the first because those triggers everything else — has been the dominant theory for 30 years. Most would consider the amyloid/Abeta cascade hypothesis to be the leading candidate for what causes Alzheimer's disease. This hypothesis has fueled most of the recent clinical trials aimed at removing Abeta/amyloids from the brain. With the failure of almost all these trials, questions have arisen as to whether deposition of amyloid and/or tau is the cause or

is a consequence of the disease [Frontiersin](#) .This is a genuinely unsettled debate in the field right now.

Tau — the second domino, and the closer link to symptoms

Tau is a protein that normally lives inside neurons and acts like scaffolding — it stabilizes the internal transport tracks (microtubules) that neurons use to move nutrients and signals around. When tau works correctly, neurons function. When it breaks down, neurons fall apart.

In Alzheimer's, tau gets abnormally phosphorylated — meaning chemical tags get attached to it that shouldn't be there. This makes tau detach from the microtubules and clump together into twisted fibers called neurofibrillary tangles (NFTs) inside the neurons. The transport system is over. The neuron starves and dies.

Of these two major AD pathological hallmarks, tau tangles have the strongest correlation with neurodegeneration and cognitive decline. [PubMed Central](#) This is critical — amyloid plaques can be present for years with little cognitive change, but once tau tangles spread, cognitive decline follows. The number of NFTs is positively correlated with the severity of AD. [Frontiers](#)

Tau also spreads through the brain in a predictable pattern — starting in the entorhinal cortex (memory hub), then spreading to hippocampus, then across the whole brain. This spreading pattern, called Braak staging, is how doctors can estimate how far the disease has progressed.

How amyloid and tau connect to each other

They are not independent. Amyloids drive tau to become toxic. We focused on the interplay and synergistic effects (**Interplay** just means two things influencing each other back and forth — not one-way, but a two-way conversation. A affects B, and B affects A in return. Synergistic **effects** means when two things combine; the damage they cause together is *greater* than the sum of what each would cause alone. $1 + 1 = 3$, not 2.), of amyloid- β and tau in the pathogenesis of AD. [Nature](#) The current understanding is amyloid accumulation comes first and creates a toxic environment that accelerates tau phosphorylation, tangle formation, and neuronal death. Tau then does the actual killing of neurons. This is why

drugs that only target amyloid have largely failed — by the time amyloid is visible, tau pathology may already be driving the damage.

APOE4 — the accelerant on top of everything

APOE is a gene that makes a protein whose primary job in the brain is lipid transport — moving cholesterol and fats between brain cells. It also helps clear amyloids. There are three versions: APOE2 (protective), APOE3 (neutral, most common), and APOE4 (the dangerous one). The difference between APOE3 and APOE4 is literally two amino acid changes at positions 112 and 158 in a 299-amino-acid protein. Those two changes cascade into enormous consequences.

Here is what APOE4 does wrong, specifically:

On amyloid: APOE4 is less effective at clearing A β due to a low removal rate of this peptide, which causes it to aggregate and lead to the formation of amyloid plaques. Additionally, APOE4 may more easily promote A β aggregation in neurotoxic oligomers and fibrils, further exacerbating amyloid pathology. [MDPI](#)

On tau: APOE4 promotes hyperphosphorylation of tau proteins, which triggers the generation of neurofibrillary tangles. APOE4 incentivizes the accumulation of these NFTs through its ability to deregulate kinases and phosphatases involved in the tau phosphorylation/dephosphorylation cycle, including glycogen synthase kinase 3 beta (GSK-3 β) and protein phosphatase 2A (PP2A). [MDPI](#)

On microglia (the brain's immune cells): APOE4 alters the morphology of microglia, causing them to have shortened and less ramified processes, impairing their ability to survey the brain for damage and clear A β and tau. [Biospective](#) On top of that, in the presence of APOE4, microglia adopt a chronic pro-inflammatory phenotype that exacerbates neuroinflammation. Microglia expressing APOE4 release high levels of pro-inflammatory cytokines, including TNF- α , IL-1 β , and IL-6. [Biospective](#)

On the blood-brain barrier: APOE4 exacerbates various properties that may influence AD risk in addition to its effects on amyloid deposition, including blood-brain barrier breakdown and neuroinflammation. [Rockefeller University Press](#) When the blood-brain barrier breaks down, toxic substances from the bloodstream leak into the brain, worsening everything.

On neurons directly: APOE4 increases tau burden in human brains and promotes the accumulation of phosphorylated tau in human neurons and mouse models. Furthermore, APOE4 increases neuroinflammation and gliosis in human AD brains. [PubMed Central](#)

On women specifically: Women carrying the APOE4 allele have a significantly higher risk of developing AD compared to men with the same genotype. This increased vulnerability is particularly noticeable after menopause, likely due to the decline in estrogen levels. Female APOE4 carriers display more severe amyloid-beta pathology, quicker cognitive decline, and a more significant accumulation of tau tangles than their male counterparts.

Things to know:

1. Synapses are the gaps between neurons where they communicate with each other.

Here's the simplest way to think about it:

Your brain has about 86 billion neurons. No two neurons actually touch each other. Between every two neurons there is a tiny gap — that gap is the synapse. Information crosses this gap using chemicals called **neurotransmitters**.

Here's the sequence of what happens:

One neuron fires an electrical signal down its length. When that signal reaches the end, it triggers the release of neurotransmitters into the gap. Those chemicals float across and bind to receptors on the next neuron, which then either fires its own signal or doesn't. That's how every thought, memory, movement, and emotion you have worked — billions of these chemical handshakes happen every second.

Why synapses matter for Alzheimer's specifically:

Synapses are the first thing that breaks down in Alzheimer's — before neurons even die. The toxic amyloid oligomers (the small clumps, before they become full plaques) attach directly to synapses and poison them. This is why memory problems are the first symptoms. Memory is literally just a pattern of synaptic connections. When those connections weaken and disappear, the memory goes with them.

Synaptic dysfunction occurs at the early stages of Alzheimer's and then progresses into neurodegeneration with neuronal cell death in later stages. [Nature](#)

So, in plain terms: synapses are where your brain actually "thinks." In Alzheimer's, amyloid breaks them first, tau kills the neurons second, and APOE4 makes both happen faster.

2. Truncation means a protein gets cut short — it's missing a piece it's supposed to have.

Think of it like this:

Every protein is basically a long chain of amino acids, like beads on a necklace. Your DNA has instructions for exactly how long that necklace should be and in what order. When truncation happens, something cuts that chain too early — so the protein ends up shorter than it's supposed to be, missing its tail end.

3. Braak staging is a system that maps how far tau tangles have spread through the brain — it's basically a roadmap of Alzheimer's progression.

They noticed that tau tangles don't appear randomly — they always spread in the same predictable order, region by region. So, they created 6 stages to describe this.

The 6 stages, plain and simple:

Stages I & II — No symptoms yet. Tau tangles appear only in the entorhinal cortex — a small region deep in the brain that acts as the gateway to memory. The person feels completely normal. This can start in your 20s or 30s in people who will eventually develop Alzheimer's decades later.

Stage III & IV — Mild symptoms begin. Tangles spread into the hippocampus — the brain's main memory-forming center. This is when people start forgetting recent events, getting lost, struggling with names. Most people get diagnosed somewhere around here.

Stages V & VI — Severe disease. Tangles spread across the entire neocortex — the outer layer of the brain responsible for language, reasoning, personality, and all higher thinking. At this point the person loses the ability to recognize family, speak coherently, or care for themselves.

Why it matters for the research:

Two things make Braak staging directly relevant to what you're studying. First, tau tangles have the strongest correlation with neurodegeneration and cognitive decline [PubMed Central](#) — meaning Braak stage predicts how sick someone actually is better than amyloid plaques do. Second, APOE4 accelerates how fast someone moves through these stages. It doesn't change the order — the order is always the same — it just makes the journey faster and more destructive.

The fact that this fixed geographical path is also one of the biggest unsolved puzzles in the field. Nobody fully understands why it always goes against the entorhinal cortex first, then hippocampus, then outward. That spreading mechanism is an active area of research right now.